



The medal points' incenter for rankings in sport

Sebastian Sitarz

Institute of Mathematics, University of Silesia, ul. Bankowa 14, 40-007 Katowice, Poland

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ABSTRACT

This work addresses the issue of finding a system of points for medals in Olympic ranking and scoring systems in sport. The incenter of a convex cone is used to obtain this system. The work deals also with the scoring systems. In applications, we present a ranking of countries in the 2010 Olympic Games and a ranking of drivers in the 2011 Formula One World Championship.

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1. Introduction

We consider a problem of building an Olympic ranking based only on medals won. Our approach is based on using points for medals. The question is: what system of points should be used? To answer this question we formulate a minimal set of conditions for the set of all reasonable points. After that we find the center of this set and the system of points for medals corresponding to this set.

The problem of building an Olympic ranking has appeared lately in some papers. Several of them base consideration on data envelopment analysis (DEA): Soares de Mello et al. [1,2] and Hai [3] use unitary input DEA models, Wu et al. [4] apply DEA to calculate the performance of the nations in the six Summer Olympic Games, and Lins et al. [5] also use DEA to build an Olympic ranking based on a country's ability to win medals in relation to its available resources. Saaty [6] proposes the analytic hierarchy process (AHP) to rank countries in the 2008 Olympic Games. Bergiante and Soares de Mello [7] present a Copeland hierarchy method for ranking countries for the 2010 Olympic Games. Sitarz [8] proposes the following points for medals: 11, 5, 2; however these points are obtained by taking into account an additional condition for the set of points—the limitation to the unit simplex.

The following parts of this work are organized as follows: Section 2 introduces the incenter of the set of points for medals. Section 3 presents the scoring systems. Applications of the approach presented are given in Section 4 for the 2010 Winter Olympic Games and the 2011 Formula One World Championship. Conclusions are discussed in Section 5.

2. The incenter of the set of points for medals

2.1. The set of points for medals

We look for a system of points for medals. We are going to formulate a minimal set of conditions generating the set of points for medals. These conditions are clear: the gold medal is assigned more points than the silver, and the silver more than the bronze. Moreover, according to Hai [3], and Soares de Mello et al. [1,2], an additional condition is needed: the

E-mail address: ssitarz@math.us.edu.pl.

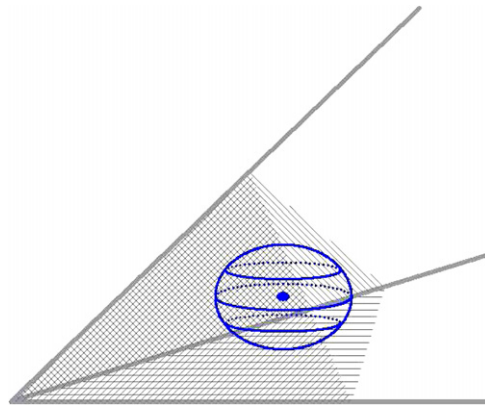


Fig. 1. The unbounded cone K with its incenter as the center of the ball.

difference between a gold and a silver medal should be bigger than the difference between a silver and a bronze medal. Thus, we consider the following set of points for medals:

$$K = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 \geq x_2 \geq x_3 \geq 0 \text{ and } x_1 - x_2 \geq x_2 - x_3\},$$

where x_1 denotes the points for gold, x_2 the points for silver and x_3 the points for bronze.

It is obvious that the above set K is a convex cone in real three-dimensional space. Moreover, the set K is unbounded.

2.2. Incenters of convex cones

A lot of multi-criteria methods and statistical methods use the mean value as a tool for aiding decisions. Similarly, we are going to use some kind of mean value to find a system of points. The mean value of a set is the center of the set. We operate on set K , so we shall find the center of set K . In the case of bounded sets the role of the center is played by a centroid. In our case, set K is an unbounded, convex cone, and thus we shall find the center of a convex cone. Such theory is analyzed by Henrion and Seeger in [9,10]. In these papers the approach most considered is finding the incenter of a convex cone. The incenter corresponds to the center of a certain largest ball inscribed in the cone K . Formally, it is defined by Henrion and Seeger [9] as an optimal solution of the following optimization problem:

$$\max_{x \in K \cap S_x} \text{dist}(x, \partial K)$$

where S_x denotes the unit sphere, ∂K denotes the boundary of set K and dist denotes the distance in the Euclidean space. Summarizing this short introduction to the centers of convex cones, it is worth noticing that this approach has good mathematical properties which were shown in papers by Henrion and Seeger [9].

2.3. Using the incenter to the set of points for medals

We are going to use the concept of the incenter to the set of points for medals. Henrion and Seeger [10] present many numerical methods to obtain the incenters for different classes of cones. By using these methods we obtain the incenter of K , which has the following form:

$$(\bar{x}_1, \bar{x}_2, \bar{x}_3) = \lambda \left((\sqrt{2} + 1)(\sqrt{3} + 2) - (\sqrt{3} + 1), (\sqrt{2} + 1), 1 \right)$$

where λ is a parameter.

The detailed computation of the above incenter is presented in the [Appendix](#). An illustration of the incenter of K is given in [Fig. 1](#).

2.4. Values of the medal points' incenter

The multiplier λ does not play a role in the rankings built by using points: $(\bar{x}_1, \bar{x}_2, \bar{x}_3)$, and thus we can consider the following triple:

$$(\bar{x}_1, \bar{x}_2, \bar{x}_3)/\lambda = \left((\sqrt{2} + 1)(\sqrt{3} + 2) - (\sqrt{3} + 1), (\sqrt{2} + 1), 1 \right) \approx (6.3, 2.4, 1).$$

The above medal points, i.e. 6.3 points for gold, 2.4 points for silver and 1 point for bronze are obtained on the basis of the approximated values. Olympic ranking is built by comparing the weighted sum of these points and the medals won (see [Section 4.1](#)).

Table 1
The scoring systems with rounded numbers.

n	Scoring positions									
	1	2	3	4	5	6	7	8	9	10
2	2.4	1								
3	6.3	2.4	1							
4	12.6	6.3	2.4	1						
5	21.4	12.6	6.3	2.4	1					
6	32.6	21.4	12.6	6.3	2.4	1				
7	46.2	32.6	21.4	12.6	6.3	2.4	1			
8	62.3	46.2	32.6	21.4	12.6	6.3	2.4	1		
9	80.9	62.3	46.2	32.6	21.4	12.6	6.3	2.4	1	
10	101.9	80.9	62.3	46.2	32.6	21.4	12.6	6.3	2.4	1

3. Scoring systems for n scored positions

3.1. The incenter in a case of n dimensions

Let us describe the generalization of the above approach. Now, we assign points not only for medals but for the first n positions. We assume that the first place is assigned more points than the second, and the second more than the third and so on. Moreover, we apply an additional condition: the difference between the i th place and the $(i + 1)$ th place should be bigger than the difference between the $(i + 1)$ th place and the $(i + 2)$ th place.

Thus, we consider the following set of points for scoring positions:

$$K = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1 \geq x_2 \geq \dots \geq x_n \geq 0 \text{ and } x_1 - x_2 \geq x_2 - x_3 \geq \dots \geq x_{n-1} - x_n\}$$

where x_1 denotes the points for the first place, x_2 the points for the second place, ..., and x_n the points for the n th place.

By using the calculation presented in the [Appendix](#) we obtain the following incenter of K :

$$\bar{x}_n = \lambda,$$

$$\bar{x}_{n-1} = \lambda(\sqrt{2} + 1),$$

$$\bar{x}_{n-2} = \lambda \left[(\sqrt{2} + 1)(\sqrt{3} + 2) - (\sqrt{3} + 1) \right],$$

$$\bar{x}_i = 3\bar{x}_{i-1} - 3\bar{x}_{i-2} + \bar{x}_{i-3}, \quad \text{for } i = n - 3, \dots, 1$$

where λ is a parameter.

3.2. Scoring systems

Let us present the scoring system based on the incenter of K . The multiplier λ does not play a role in the rankings obtained by using points generated by $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{n-1}, \bar{x}_n)$; thus we can consider the following values: $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{n-1}, \bar{x}_n)/\lambda$. [Table 1](#) presents the rounded up coefficients of the above vector. As we can see in [Table 1](#), the points for positions are constant for the last positions, which means that the last scoring position always scores 1 point, the one before the last scoring position always scores 2.4 points and so on.

We can use the points from [Table 1](#) to build the ranking in the case of scoring n positions. For example, if ten positions score, then we use points from the 10th row of [Table 2](#): 101.9 points for the first position, 80.9 points for the second position and so on. The ranking is built by comparing the weighted sum of these points and the positions (see [Section 4.2](#)).

4. Applications

4.1. The 2010 Winter Olympic Games in Vancouver

We consider the 2010 Winter Olympic Games for the top ten countries. [Table 2](#) presents the medal table with countries ordered by lexicographic ranking (so called “gold first” ranking). In [Table 2](#) the column “Sum of incenter points” presents a weighted sum of points $\bar{x}/\lambda$ and medals (for the values of $\bar{x}/\lambda$ see [Section 2.4](#)). The last column “Incenter ranking” of [Table 2](#) shows the positions of countries in the ranking obtained by using the values from the column “Sum of incenter points”. As we can see in [Table 2](#) the lexicographic ranking and the “incenter ranking” differ. The reason is that the incenter approach takes into account the mean value for medals, not only gold medals won.

4.2. The 2011 Formula One World Championship

The problem of the Formula One system of points has been studied in the following papers: Langen and Krauskopf [[11](#)], Haigh [[12](#)], Soares de Mello et al. [[13](#)], Kladraba [[14](#)]. Here we adopt the incenter method. We consider the 2011 Formula One

Table 2

Olympic ranking by the “gold first” method and a new ranking obtained by using the incenter.

	Country	Gold	Silver	Bronze	Sum of incenter points	Incenter ranking
1	Canada	14	7	5	109.8	1
2	Germany	10	13	7	101.2	3
3	USA	9	15	13	105.7	2
4	Norway	9	8	6	81.8	4
5	South Korea	6	6	2	54.2	5
6	Switzerland	6	0	3	40.7	7
7	China	5	2	4	40.2	8
7	Sweden	5	2	4	40.2	8
9	Austria	4	6	6	45.6	6
10	Netherlands	4	1	3	30.5	11
11	Russia	3	5	7	37.9	10
12	France	2	3	6	25.8	12
13	Australia	2	1	0	15.0	15
14	Czech Republic	2	0	4	16.6	13
15	Poland	1	3	2	15.5	14
16	Italy	1	1	3	11.7	16
17	Belarus	1	1	1	9.7	17
17	Slovakia	1	1	1	9.7	17
19	Great Britain	1	0	0	6.3	20
20	Japan	0	3	2	9.2	19
21	Croatia	0	2	1	5.8	22
21	Slovenia	0	2	1	5.8	22
23	Latvia	0	2	0	4.8	24
24	Finland	0	1	4	6.4	21
25	Estonia	0	1	0	2.4	25
25	Kazakhstan	0	1	0	2.4	25

Table 3

The official ranking and a new ranking obtained by using the incenter.

	Driver	Positions (1st, 2nd, ..., 10th)	Official sum	Sum of incenter points	Incenter ranking
1	Vettel	(11, 5, 1, 1, 0, 0, 0, 0, 0, 0)	392	1634.1	1
2	Button	(3, 4, 5, 2, 0, 3, 0, 0, 0, 0)	270	1097.5	2
3	Webber	(1, 2, 7, 6, 2, 0, 0, 0, 0, 0)	258	1042.6	3
4	Alonso	(1, 5, 4, 4, 2, 1, 1, 0, 0, 0)	257	1039.7	4
5	Hamilton	(3, 3, 0, 5, 2, 1, 1, 1, 0, 0)	227	884.9	5
6	Massa	(0, 0, 0, 0, 6, 5, 2, 1, 1, 0)	118	336.0	6
7	Rosberg	(0, 0, 0, 0, 2, 4, 5, 1, 1, 1)	89	223.2	7
7	Schumacher	(0, 0, 0, 1, 3, 2, 1, 2, 2, 0)	76	216.6	8
9	Sutil	(0, 0, 0, 0, 0, 2, 2, 2, 3, 0)	42	87.7	11
10	Petrov	(0, 0, 1, 0, 1, 0, 0, 1, 3, 2)	37	110.4	9
11	Heidfeld	(0, 0, 1, 0, 0, 0, 1, 3, 0, 1)	34	94.8	10
12	Kobayashi	(0, 0, 0, 0, 1, 0, 2, 0, 2, 4)	30	66.6	12
13	Di Resta	(0, 0, 0, 0, 0, 1, 1, 2, 1, 3)	27	51.9	13
14	Alguersuari	(0, 0, 0, 0, 0, 0, 2, 3, 0, 2)	26	46.0	14
15	Buemi	(0, 0, 0, 0, 0, 0, 0, 2, 2, 3)	15	20.4	16
16	Perez	(0, 0, 0, 0, 0, 0, 1, 1, 1, 2)	14	23.3	15
17	Barrichello	(0, 0, 0, 0, 0, 0, 0, 0, 2, 0)	4	4.8	17
18	Senna	(0, 0, 0, 0, 0, 0, 0, 0, 1, 0)	2	2.4	18
19	Maldonado	(0, 0, 0, 0, 0, 0, 0, 0, 0, 1)	1	1.0	19

World Championship for the top ten drivers. Table 3 presents the official ranking of drivers, obtained by using the points for the top ten positions (according to the Federation Internationale de l'Automobile (FIA)): (25, 18, 15, 12, 10, 8, 6, 4, 2, 1). The sum of these points and positions is given in the column “Official sum” in Table 3. We take the alternative points for positions on the basis of the 10th row of Table 1—the points obtained by the incenter approach. The weighted sum of these points and positions is given in the column “Sum of incenter points” in Table 3. The last column “Incenter ranking” of Table 3, presents the positions of drivers in the ranking obtained by using the values from the column “Sum of incenter points”. As we can see in Table 3, the official ranking and the “incenter ranking” do not differ much. The reason is clear: the points for positions of these two systems differ.

5. Summary

We focused on the problem of building an Olympic ranking based only on medals won. We used points for medals. We formulated a minimal set of conditions for points and we obtained the convex cone K representing points for medals. We applied the concept of incenters for convex cones. After that we found the incenter of the set K and the system of medal

points corresponding to this set. We presented applications in the form of a ranking of countries in the 2010 Olympic Games and a ranking of drivers in the 2011 Formula One World Championship.

Let us note some properties of our approach. We operated with the minimal set of conditions for the points. We used the concept of the center of a convex cone—a generalization of the centroid of a bounded set. In other words, the center of the set was taken to use the mean value approach, which is often practised in statistical and multi-criteria problems. Moreover, we were able to use our approach for building the scoring systems in the case of more than three positions scoring. In future, it will be worth considering the usage of sensitivity analysis for the ranking by multi-objective methods; see Sitarz [15,16].

Appendix

Below, we present how to compute the incenter of the cone K :

$$K = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1 \geq x_2 \geq \dots \geq x_n \geq 0 \text{ and } x_1 - x_2 \geq x_2 - x_3 \geq \dots \geq x_{n-1} - x_n\}.$$

The above cone K can be written in the simpler form

$$K = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1 - x_2 \geq x_2 - x_3 \geq \dots \geq x_{n-1} - x_n \geq 0 \text{ and } x_n \geq 0\}.$$

By the definition of the incenter given in Section 2.2, we obtain the following optimization problem:

$$\begin{aligned} \max \min & \left\{ \frac{x_1 - 2x_2 + x_3}{\sqrt{6}}, \frac{x_2 - 2x_3 + x_4}{\sqrt{6}}, \dots, \frac{x_{n-2} - 2x_{n-1} + x_n}{\sqrt{6}}, \frac{x_{n-1} - x_n}{\sqrt{2}}, x_n \right\} \\ & x_1 - 2x_2 + x_3 \geq 0, x_2 - 2x_3 + x_4 \geq 0, \dots, x_{n-2} - 2x_{n-1} + x_n \geq 0, x_{n-1} - x_n \geq 0, \\ & x_n \geq 0, \\ & x_1^2 + x_2^2 + \dots + x_n^2 = 1. \end{aligned}$$

By using the conditions for an optimal solution, we obtain that an optimal solution $\bar{x}$ satisfies

$$\frac{x_1 - 2x_2 + x_3}{\sqrt{6}} = \frac{x_2 - 2x_3 + x_4}{\sqrt{6}} = \dots = \frac{x_{n-2} - 2x_{n-1} + x_n}{\sqrt{6}} = \frac{x_{n-1} - x_n}{\sqrt{2}} = x_n.$$

To compute the optimal solution we use an iterative method:

$$\begin{aligned} \bar{x}_n &= \lambda, \bar{x}_{n-1} = \lambda(\sqrt{2} + 1), \bar{x}_{n-2} = \lambda \left[(\sqrt{2} + 1)(\sqrt{3} + 2) - (\sqrt{3} + 1) \right], \\ \bar{x}_i &= 3\bar{x}_{i-1} - 3\bar{x}_{i-2} + \bar{x}_{i-3}, \quad \text{for } i = n-3, \dots, 1 \end{aligned}$$

where λ is a parameter.

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